DC Fast Charger Controller, EV–Charger Communication Protocols, Charger-Side Hardware, and Buck / Boost Converters

A Complete Technical Research & Interview-Preparation Report

Beginner → Intermediate Engineering Level

Prepared for: Technical interview preparation — automotive / EV engineering startup

Topics as advised by the engineering manager: DC fast charger controller · EV ↔ charger communication · charger-side hardware · protocol internals · buck & boost converters

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All diagrams are original drawings created for this report.

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How to read this report. Yellow italic boxes give short **Kannada** explanations of the hardest concepts. Green boxes are **verified facts with citations**; blue boxes flag **architecture-dependent** details; red boxes flag **safety-critical or commonly-confused** points. Illustrative example calculations always state their assumptions.

1. Executive Summary

1.1 What is a DC fast charger?

A **DC fast charger** (also called DCFC, or "Level 3" informally in the US) is an off-board charging station that converts grid AC power into DC power *inside the charger itself* and feeds that DC directly into the vehicle's high-voltage (HV) battery. This is possible only because the charger takes over the job that the vehicle's on-board charger (OBC) normally does during AC charging: rectification, power-factor correction, galvanically-safe voltage conversion and current regulation. Because the power electronics are off-board, they can be sized far larger — commercial DC fast chargers typically deliver roughly 50 kW to 350+ kW per dispensing unit (architecture-dependent), versus roughly 3–22 kW for typical AC charging [Source: NREL/DOE infrastructure reports, §17].

> *(ಕನ್ನಡ: DC ಫಾಸ್ಟ್ ಚಾರ್ಜರ್ ಎಂದರೆ — ಕರೆಂಟ್ ಅನ್ನು ಚಾರ್ಜರ್ ಒಳಗೇ AC ಯಿಂದ DC ಆಗಿ ಪರಿವರ್ತಿಸಿ, ನೇರವಾಗಿ ಕಾರಿನ ಬ್ಯಾಟರಿಗೆ ಕೊಡುವ ಯಂತ್ರ. ಕಾರಿನ ಒಳಗಿನ ಚಿಕ್ಕ ಚಾರ್ಜರ್ ಬದಲು ದೊಡ್ಡ ಹೊರಗಿನ ಚಾರ್ಜರ್ ಈ ಕೆಲಸ ಮಾಡುತ್ತದೆ — ಆದ್ದರಿಂದ ವೇಗವಾಗಿ ಚಾರ್ಜ್ ಆಗುತ್ತದೆ.)*

1.2 Why is a charger controller required?

The power hardware alone is "blind". A **charger controller** (an embedded MCU/SoC system) is required because the whole session is a *negotiated, closed-loop, safety-supervised* process. The controller must:

- **Communicate** with the EV's battery management system (BMS) over the CCS digital protocol, and with the vehicle's analog Control Pilot (CP) signalling, to agree exactly what voltage and current may be delivered, at every instant.
- **Regulate** the power stage in closed loop: sample output voltage/current via ADC, adjust the PWM of the DC/DC modules, and track the EV's request within tolerance.
- **Protect**: supervise insulation resistance, connector temperature, DC-link voltage, coolant temperature, contactor feedback, and execute safe shutdowns.
- **Manage the session**: authorise the user, meter energy, talk to the payment/back-end server (OCPP), and log everything for diagnostics.

IEC 61851-23 makes this supervision a certification requirement for DC EVSE, including insulation monitoring, pre-charge control and emergency shutdown paths [Source: Bender, Gongyuan IEC 61851-23 guides — §17].

1.3 AC charging vs DC fast charging

	AC charging (Mode 3, Type 2)	DC fast charging (Mode 4, CCS2)
Converter location	On-board charger (OBC) *inside the EV*	Off-board, *inside the charger*
What flows in the cable	AC mains voltage	Already-rectified DC (up to ~1000 V systems — architecture-dependent)
Typical power	~3–22 kW (limited by OBC)	~50–350+ kW (architecture-dependent)
Who sets charge rate	OBC (charger only signals available current on CP duty cycle)	EV BMS requests exact V/I; charger follows in closed loop
Communication	CP PWM only (IEC 61851-1); optional ISO 15118 for smart functions	CP PWM (5% = digital signalling) **plus** PLC data link: DIN 70121 / ISO 15118-2/-20
Battery-side switching	OBC output connects after internal contactors	Charger output must be pre-charged to battery voltage before HV contactor closure

Key standards: IEC 61851-1 (conductive charging, CP/PP, basic signalling) [Source: eInfochips IEC 61851 guide; Geocadra IEC 61851 summary — §17]; IEC 61851-23 (DC EVSE safety); DIN SPEC 70121 and ISO 15118-2/-20 (high-level DC communication); CCS = Combined Charging System combining these [Source: CharIN HPC guide — §17].

1.4 Power stage vs control/communication stage

Every DC fast charger is really **two machines in one box**:

- **Power stage** — the "muscle": breakers, EMI filter, PFC rectifier, DC-link capacitors, isolated DC/DC modules, contactors, cabling. It handles hundreds of amperes. It does not think; it obeys.
- **Control/communication stage** — the "brain and nervous system": MCU(s), gate drivers (the nerve endings), sensors (the senses), PLC modem + SECC (the translator that talks to the EV), HMI/RFID, and backend connectivity. It senses, decides, and commands.

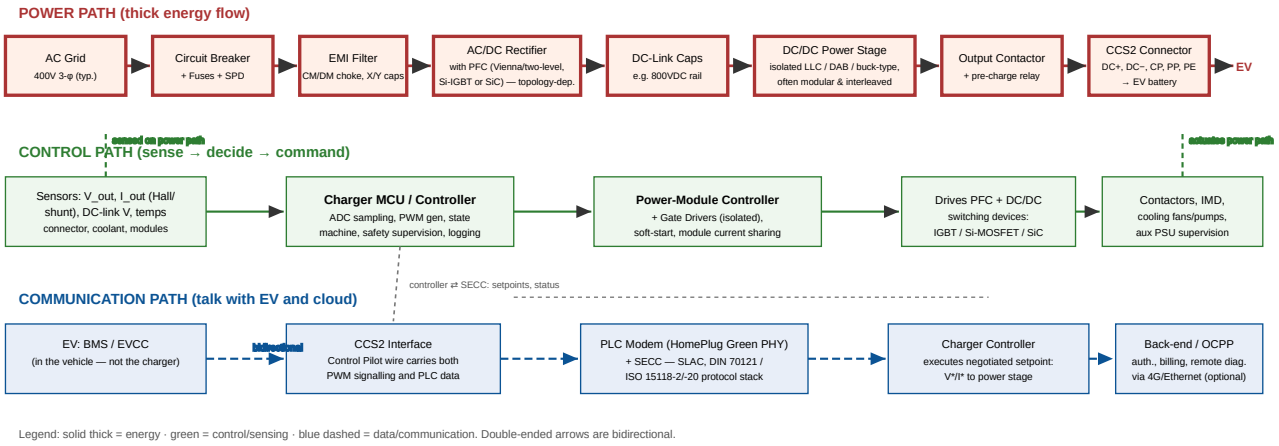
The interview one-liner: *"The power stage converts and delivers energy; the controller converts information — sensing the EV's request and enforcing safety — into precise PWM commands for that power stage."*

1.5 The complete system in simple language

A driver plugs in. The charger notices the plug (Proximity Pilot) and the car (Control Pilot voltage change), locks the connector, and switches CP to 5% PWM to say "I can speak digitally". A data modem conversation (SLAC over the same CP cable) builds an Ethernet-over-powerline link. Car and charger exchange capabilities, the charger checks insulation, matches its output voltage to the battery (pre-charge), closes its contactors, and then delivers exactly the current the BMS asks for, re-checked ~10 times per second, with both sides ready to stop within milliseconds if anything looks wrong. When the BMS says stop, the charger ramps down, opens contactors, unlocks, and sends the billing record. That is the whole story this report unpacks.

2. Complete DC Fast Charger Architecture

Fig. 1 — DC Fast Charger: Power Path, Control Path, Communication Path



Legend: solid thick = energy · green = control/sensing · blue dashed = data/communication. Double-ended arrows are bidirectional.

Block-by-block explanation (power path)

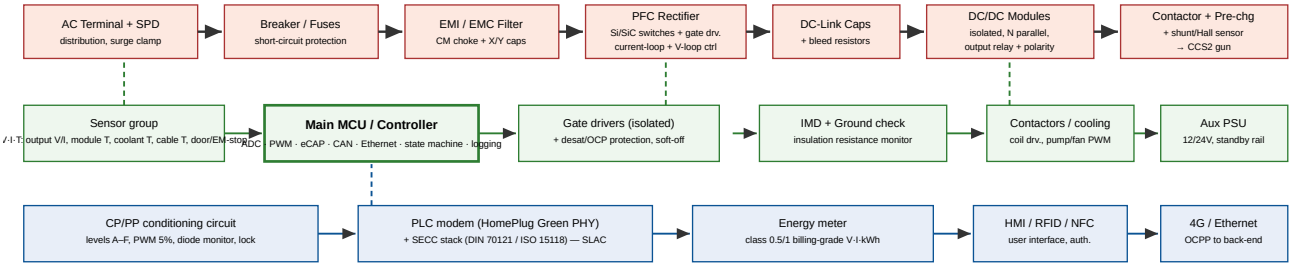
Block	Purpose	Inputs → Outputs	Interaction with neighbours	Type
AC input / distribution	Safe entry of grid energy, phase distribution	Grid 3-φ AC → terminal busbars	Feeds breaker; SPG/earth stud reference	Essential
Breaker / fuses / SPD	Overcurrent & surge isolation; lightning/grid-transient clamp	AC in → protected AC out	Trips on fault; SPD status wired to MCU	Essential
EMI filter	Blocks switching noise from entering/leaving the grid (EMC)	Noisy AC → clean AC	Sits between grid and PFC; heatsink monitored	Essential (IEC 61851-23 EMC)
PFC rectifier (AC/DC)	Converts AC to DC while forcing input current sinusoidal (PF≈0.99)	3-φ AC → regulated DC-link (e.g. 800 V)	Gate-driven by controller; feeds DC-link caps	Essential
DC-link capacitors	Energy buffer; absorbs PFC ripple, supplies DC/DC stage	Charged rail → stiff DC bus	Voltage sensed by MCU; discharge circuit safety-faults it	Essential
DC/DC stage	Isolated voltage conversion, output V/I regulation (the "engine")	DC-link → regulated EV-side DC	PWM via gate drivers; parallel/interleaved modules	Essential (topology varies)
Contactors + pre-charge	Connect/disconnect EV safely; limit inrush into cable+caps	DC bus → CCS2 pins	Coil drivers + auxiliary feedback contacts to MCU	Essential
CCS2 outlet	Physical interface + CP/PP signalling + PLC coupling	DC+, DC−, CP, PP, PE	Cable temp sensed; lock solenoid driven	Essential

Figure 1 — Original overall architecture diagram. Power path (red), control path (green), communication path (blue). Topology names (Vienna, LLC, DAB) are typical examples, not universal — every such choice is architecture-dependent.

> **Architecture-dependent notes (Fig. 1).** Many designs place the PFC and DC/DC stages in a separate power cabinet with their own module controllers; the "Charger MCU" may then be a hierarchy (site controller + power-module controllers + dispenser controller) connected by CAN or Ethernet. Some 400 V-class chargers use non-isolated buck-derived output stages; most modern high-power units use isolated stages (LLC, DAB, or CLLC). Liquid cooling is standard above ~150 kW; air-cooled below. Do not present any single topology as "the" topology in the interview.

3. Charger-Side Hardware and Components

Fig. 2 — Charger-Side Hardware Block Diagram (one plausible architecture)



Red = energy chain · Green = control & safety chain · Blue = user & data chain. Arrows into the MCU are ADC/digital inputs; arrows out are PWM/GPIO/drive signals.

Typical signals between blocks (illustrative, verify against your product's datasheet)

Link	Signal	Typical form / rating (illustrative)
Sensor → MCU	Output voltage feedback	0–5 V or 0–10 V analog into ADC; isolated (AMC/ISO amp) for HV side
MCU → gate drivers	PWM	3.3 V logic, 16–100 kHz (PFC/DC-DC dependent); isolated gate drive ±15 V typical for SiC
MCU ⇄ SECC	EV request / charger limit	CAN or SPI/Ethernet between SECC host and controller (vendor-specific)
IMD → MCU	Insulation status	Digital OK/warn/fault or serial; threshold per IEC 61851-23 (warning vs fault levels)

Figure 2 — Original charger-side hardware diagram. One concrete wiring of the blocks from Fig. 1; real products redistribute these functions across boards and cabinets.

3.1 Master component table

Columns: **Function · I/O · Signals/ratings · MCU link · Failure modes · Test method · Status**. Ratings marked (ill.) are illustrative industry-typical values, not specification claims; actual ratings depend on the manufacturer and power class.

#	Component	Main function	Input → Output	Typical signals / ratings (ill.)	Connection to MCU	Common failure modes	Testing / debugging	Status
1	AC input terminal & power distribution	Safe grid entry; phase/busbar distribution	Grid 3-φ → internal bus	400 V AC 3-φ typical; M6–M10 studs	Torque/thermistor check only	Loose lug, thermal runaway at joint	Thermal camera under load, torque check	Essential
2	Circuit breakers & fuses	Overcurrent / short-circuit isolation	AC in → protected out	MCCB sized to site feed; fuses for semiconductors	Aux trip contact → GPIO	Nuisance trip, welded contacts, fuse fatigue	Trip test, contact resistance measurement	Essential
3	Surge Protection Device (SPD)	Clamps lightning/grid transients	Parallel across AC lines → PE	MOV based; Class I/II per IEC 61643	Replacement-window status flag → GPIO	MOV end-of-life, thermal disconnect open	Visual window, status contact, leakage test	Essential (site-dependent class)
4	EMI/EMC filter	Attenuate conducted EMI both directions	Noisy AC → clean AC	CM chokes, X/Y capacitors	Temperature (optional)	Capacitor failure, choke saturation	LISN scan vs IEC 61851-23 limits; hipot for Y-caps	Essential
5	AC/DC rectifier	Grid AC → DC bus	AC → DC-link	Diode/active front end; output e.g. 800 V DC (ill.)	Gate PWM + V/I feedback	Switch short, gate-drive loss, dc ripple rise	Scope on gate waveforms, ripple analysis	Essential
6	PFC (function of #5)	Force sinusoidal grid current, PF≈0.99	AC current shaped by PWM	THDi targets <5% at rated (design goal, ill.)	Inner current loop + outer V loop in MCU/DSP	Loop instability, PF collapse under light load	Power analyser PF/THDi sweep vs load	Essential (in active front end)
7	DC-link capacitors	Buffer energy, stabilise bus	Charged rail → stiff DC	Electrolytic or film; bleed resistors	Voltage sense → ADC; discharge control	Drying (electrolytic), swelling, capacity fade	ESR meter, ripple current thermography	Essential
8	DC/DC converter / power modules	Isolated, regulated output conversion	DC-link → EV-side DC	LLC / DAB / CLLC modules, paralleled	Per-module enable, PWM, current-share bus	Transformer insulation, resonant cap stress	Efficiency map, full-load burn-in	Essential
9	MOSFET / IGBT / SiC MOSFET	Switching elements	—	Si IGBT 1200 V class common; SiC offers lower switching loss at high f (vendor docs)	via gate drivers	Short-circuit from avalanche/overload; gate-oxide	$R_{DS(on)}$/$V_{CE(sat)}$ trending, thermal scan	Essential
10	Gate-driver circuits	Level-shift & isolate MCU PWM to power switch	3.3/5 V logic → ±gate volts	Optical/digital isolation; Miller clamp; desat detect	PWM out; fault-in	Bootstrap cap degradation, isolation crack	Scope gate voltage at device; desat trip test	Essential
11	Voltage sensors	Measure DC output / DC-link voltage	HV → scaled analog/digital	Isolated amplifiers or resistive dividers + isolation	ADC channel	Drift, gain error, broken sense line	Calibration vs calibrated DMM across range	Essential

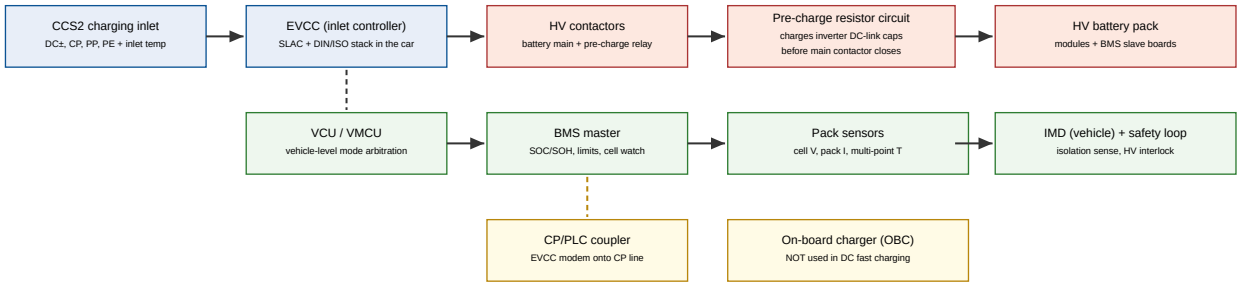
12	Current sensors (shunt/Hall)	Measure DC output current	Busbar current → signal	Shunt+mV iso-amp; closed-loop Hall; fluxgate (highest accuracy)	ADC channel	Shunt solder fatigue, Hall offset drift	Inject known current; compare vs meter	**Essential**
13	Temperature sensors	Monitor modules, coolant, cable, connector	—	NTC/PT100, thermocouples	ADC (with pull-up reference)	Probe detachment, connector corrosion	Ice-bath / reference comparison; plausibility check vs twin sensors	**Essential**
14	MCU / embedded controller	Brain: loops, state machine, comms, logging	Sensors in, drives out	Industrial MCU/DSP class, redundant safety paths common	— (is the hub)	Watchdog reset, flash corruption, brown-out	JTAG/trace, fault-injection bench	**Essential**
15	ADC + PWM peripherals	Digital link between analog world and control code	Analog in → samples; timer → pulses	12–16-bit ADC; ePWM with dead-time insertion	On-chip	Reference drift, missed samples	Known-input calibration; dead-time scope check	**Essential**
16	Auxiliary power supply	Powers control electronics from grid, incl. standby	AC → 12/24/5/3.3 V rails	Industrial flyback/QR modules; hold-up for safe shutdown	Health flags, sequencing GPIOs	Capacitor aging, latch-up after surge	Rail Ripple & brown-out test, hold-up timing	**Essential**
17	Contactors + drivers	Mechanically connect/disconnect DC output	DC bus → gun	DC-rated contactors (e.g. 500–1000 V DC class, ill.); coil 12/24 V	Driver + aux contact feedback (weld detection)	Contact welding, chatter, coil burnout	Aux-contact vs coil-command consistency test	**Essential**
18	Pre-charge circuit	Limit inrush into EV-side capacitance via resistor/relay	Bus → RC network → gun	Resistor sized for energy per session	Relay GPIO; V-match comparison in software	Resistor fatigue after many cycles	Measure inrush V/t curve; count energy budget	**Essential**
19	Discharge circuit	Bleed output to safe voltage after stop	Gun terminals → bleed	Active or resistive; target "safe" threshold per standard	Commanded after contactor open	Bleed resistor open (dangerous live plug)	Discharge-time measurement after session	**Essential**
20	Insulation Monitoring Device (IMD)	Detects low isolation between DC lines and earth	DC+ / DC− vs PE	Warning and fault thresholds; test before & during session per IEC 61851-23 practice	Status/serial to MCU	False positives in humid conditions	Resistor-bank injection test	**Essential**
21	Ground/earth monitoring	Verify protective earth integrity before energising	PE loop	Resistance threshold	Digital status	Corroded earth stud	Loop resistance meter	**Essential**
22	PLC modem + SECC	High-level comms with EV over CP line	CP line ⇄ Ethernet/IP	HomePlug Green PHY; SLAC; IPv6/TCP(-TLS)	Ethernet/SPI to host controller	No SLAC match, noisy CP coupling	SLAC sniffer, protocol analyser logs	**Essential** (CCS)

23	Control Pilot (CP) interface	Analog state + PWM signalling	CP pin ⇄ MCU	±12 V levels, 1 kHz PWM; 5% = digital (CCS)	Comparator + ADC + PWM gen	Diode monitor fail, level shift drift	Oscilloscope on CP; state A–F sequence test	**Essential**
24	Proximity Pilot (PP) interface	Detect plug insertion & cable current rating	PP pin → resistor divider	Resistance-coded (values differ AC vs CCS; check standard)	ADC / GPIO	Bent pin, wrong cable resistor	Resistance measurement vs table	**Essential**
25	CCS2 connector & cable	Physical + signalling interface to EV	DC±, CP, PP, PE	Liquid-cooled cables at high power (ill.); inlet temp sensor	Temp sensor + lock solenoid feedback	Pin wear/pitting, cable damage, lock jam	PIN contact-resistance check, lock cycle test	**Essential**
26	Cooling: fans, pumps, coolant, thermal sensors	Remove heat from modules & cable	—	Flow & level switches, pump tach	PWM drive + flow feedback	Pump wear, leaks, clogged filters	Flow/ΔT test, coolant quality check	**Architecture-dependent** (liquid vs air)
27	Energy meter	Billing-grade energy measurement	V·I samples → kWh	Class 0.5/1 (ill.)	Modbus/UART pulse out	Calibration drift	Field calibration verifier	**Essential** for billing
28	HMI / display	User status, instructions, idle screen	—	Touch/LED	Serial/USB/LVDS	Touch failure, sunlight washout	Boot & touch test rig	**Optional/varies**
29	RFID / NFC authentication	Driver identification	Card read → ID	13.56 MHz ISO 14443/A (ill.)	UART/USB reader	Antenna detune	Card read range test	**Optional** (P&C can replace)
30	Ethernet, CAN, RS-485, cellular	Internal & external data links	—	CAN between sub-controllers; 4G to back-end	Peripheral / module	Cable EMC issues, SIM expiry	Bus traffic capture, RSSI logs	**Varies**
31	Emergency stop & interlocks	Immediate safe shutdown on operator action	—	Hardwired E-stop chain, door switches	Interrupt + hardware latch	Contact oxide (rare use), wiring break	Monthly press test + contact chain measurement	**Essential**
32	Backend / OCPP connectivity	Central management, auth, billing	—	OCPP 1.6 / 2.0.1 over WebSocket (OCA docs)	App-layer on main controller	Server cert expiry, firewall blocks	OCPP conformance logs	**Optional / networked models**

> **Design variations to mention in the interview.** (1) Integrated "all-in-one" chargers vs split power-cabinet + dispenser architectures. (2) Air-cooled ≤150 kW vs liquid-cooled ≥150 kW-class designs. (3) OCPP is charger ↔ *server* — it never talks to the EV; the EV always speaks CCS/DIN/ISO over CP+PLC. (4) Some regions/utilities add DC meters vs AC-metering-with-loss-factor models. (5) Plug & Charge (ISO 15118) can remove the need for RFID entirely.

4. EV-Side Hardware

Fig. 3 — EV-Side Hardware (what is inside the vehicle during DC charging)



Blue = communication chain · Red = power chain · Green = sensing/decision chain · Yellow = context blocks. The OBC is bypassed entirely during DC fast charging.

Figure 3 — Original EV-side hardware diagram. Note the EVCC implements the same DIN 70121/ISO 15118 stack the SECC implements in the charger.

4.1 EV-side components explained

- **CCS2 charging inlet** — the combined AC + DC socket; exposes DC+, DC−, CP, PP, PE. Includes inlet temperature sensing and a mechanical lock actuator driven by the vehicle.
- **BMS (Battery Management System)** — measures every cell group's voltage, pack current and many temperature points; computes state of charge (SOC) and state of health (SOH); decides the *maximum* permissible charge voltage/current at each moment; commands pack contactors. During DC charging it is the **authoritative** decision-maker for what the battery accepts.
- **VCU (Vehicle Control Unit)** — arbitrates vehicle modes (drive, AC charge, DC charge); authorises the EVCC to begin a session; coordinates thermal preconditioning.
- **HV battery pack** — series cell modules; its total voltage is what the charger output must be matched to during pre-charge.
- **Pack sensors** — cell voltages (per module), pack current (Hall/shunt), distributed temperatures; inputs to BMS limits.
- **HV contactors + pre-charge** — the pack stays disconnected until (a) the charger output voltage ≈ pack voltage, and (b) the vehicle's own pre-charge has filled its inverter capacitors; this prevents destructive inrush.
- **Insulation monitoring & safety systems** — the EV continuously monitors HV isolation and the interlock loop (HVIL); any breach opens contactors.
- **EVCC (EV Communication Controller)** — the vehicle-side counterpart of the SECC: runs SLAC as "peer discovery", then the DIN/ISO application protocol; connects via a PLC modem coupled onto the CP line.

4.2 Who owns which function?

Function	Owner	Notes
Rectification, PFC, isolation, output regulation	Charger	Entire power stage is off-board in DC charging
Insulation check of the DC output circuit	Charger (IMD)	IEC 61851-23 practice; result also exchanged in CableCheck
Pre-charge voltage matching	Charger adjusts output; EV verifies	Confirmed in PreCharge message exchange
Deciding charge current/voltage limits	**EV (BMS)**	BMS computes from cell data; charger must comply and also enforce its own limits (min of both)

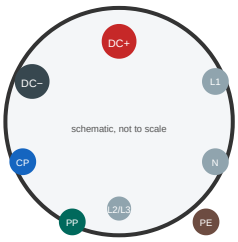
Connector lock	Vehicle locks, per standard family; charger respects lock state	Architecture/version-dependent detail
Session authorisation / billing	Charger + back-end (OCPP); EV cert with Plug & Charge	ISO 15118-2 adds P&C; DIN 70121 has no P&C
Cell-level protection, SOC/SOH	EV (BMS)	Never the charger's job — it cannot see cells

> **Common interview trap.** "The charger talks directly to the BMS." False in general: the charger's SECC talks to the **EVCC**; the EVCC relays to the BMS over vehicle-internal networks (e.g. CAN), using OEM-specific internal messages. The charger never sees raw cell data.

5. EV ↔ Charger Communication: Complete Protocol Explanation

5.1 CCS2 connector pins and their purposes

Fig. 4 — CCS2 (Combo 2) Connector Pins and Functions



Pin	Purpose	Used in DC session?
DC+ / DC−	DC energy transfer to battery	Yes — main power path
CP (Control Pilot)	Analog PWM states A–F; 5% PWM = digital; carries HLC	Yes — dual role (analog + PLC)
PP (Proximity Pilot)	Plug presence + cable/connector rating coding (resistance-coded)	Yes — latch detection
PE (Protective Earth)	Safety earthing; reference for IMD measurements	Yes
L1 / N / L2/L3 (AC pins)	AC charging via same combo inlet; unused for DC-only mode	No — de-energised

Pin placement is schematic only. CP carries BOTH the analog PWM and the PLC data signal (superimposed via modem coupling).

Figure 4 — Original CCS2 pin/function illustration.

5.2 CP and PP — the physical layer

Control Pilot (CP): a 1 kHz PWM signal between charger and EV defines states and available current. Verified state table [Source: Geocadra IEC 61851 summary; eInfochips IEC 61851 guide — §17]:

CP state	CP voltage	Meaning
A	+12 V DC	Standby — no vehicle connected
B	+9 V (PWM)	Vehicle detected, not ready
C	+6 V (PWM)	Ready / charging — power contacts closed
D	+3 V (PWM)	Ventilation required (legacy)
E	0 V	No power (error state)
F	−12 V	Pilot fault

Duty-cycle coding of available current (AC case) per IEC 61851-1: values such as 10%–85% map linearly to current limits in defined bands, and **5% duty cycle signals "digital communication required"** — the case used for CCS DC charging [Source: Pico forum technical note; nvdungx charging-standard summary — secondary sources, flagged as such; consult IEC 61851-1 for normative tables].

Fig. 5 — Control Pilot: circuit concept and PWM levels

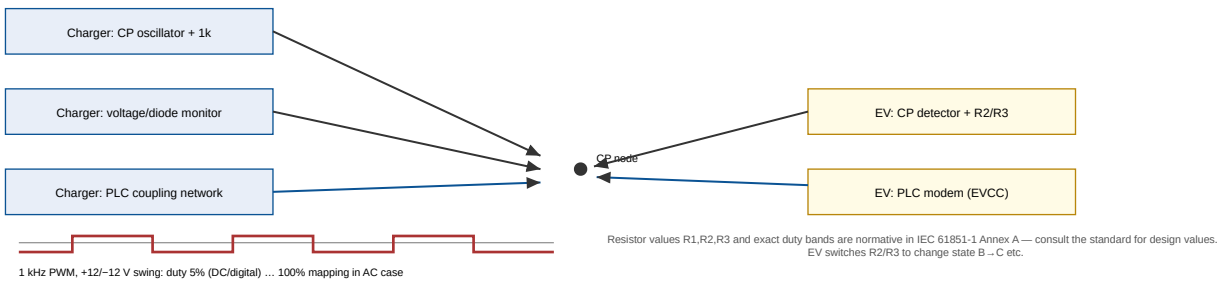


Figure 5 — Original CP/PWM concept diagram (simplified educational schematic; not a normative circuit).

5.3 PLC, HomePlug Green PHY, SLAC

Power Line Communication (PLC): for CCS DC charging, digital data is transmitted over the CP line as an RF signal using the **HomePlug Green PHY** standard — a low-power variant of HomePlug AV, using OFDM in the 2–30 MHz band, coexisting with the analog PWM on the same wire [Source: ISO 15118-3 framework; Codico HPGP module page — §17].

SLAC (Signal Level Attenuation Characterisation): the link-establishment procedure from ISO 15118-3. The EV (EVCC) transmits "sounding" messages; the charger (SECC) measures the signal attenuation on its own CP line and matches to the plug whose attenuation profile corresponds to *its* connector — so each gun pairs with exactly one car even at a multi-stall site. The sequence CM_SLAC_PARM, CM_START_ATTEN_CHAR, CM_MNBC_SOUNDING, CM_ATTEN_CHAR, CM_SLAC_MATCH ends with both sides deriving the same NMK (network membership key) and session MAC [Source: ISO 15118-3 sequence chart — FHWA docket attachment; zeropath SLAC summary — §17].

5.4 EVCC, SECC, DIN 70121, ISO 15118

Term	Meaning	Role
EVCC	EV Communication Controller	Vehicle-side protocol stack + PLC modem controller
SECC	Supply Equipment Communication Controller	Charger-side protocol stack + PLC modem controller
DIN SPEC 70121	German precursor spec (2011-ish)	First widely-deployed **DC** charging protocol; **no Plug & Charge, no TLS** — external authentication via screen/RFID
ISO 15118-2	International standard (2014)	AC+DC messaging; adds Plug & Charge (certificate-based auth) and optional TLS; XML/EXI encoded over IP
ISO 15118-20	Generation 2 (2022)	Bidirectional power transfer (V2G/V2H), improved security/TLS mandates, TCP/IPv6, new DC message set [Source: iTeh EN ISO 15118-20 entry; EVRange; elintacharge — §17]

Transport stack (educational summary): **CP line (physical, PLC-coupled) → HomePlug Green PHY MAC/PHY → IPv6 → TCP (+TLS where used) → V2GTP/EXI-encoded application messages (DIN 70121 / ISO 15118-2/-20)** [Source: Typhoon HIL ISO 15118 documentation — §17].

5.5 Message flow and negotiation

Verified ISO 15118-2 DC message sequence [Source: Typhoon HIL ISO 15118-2 protocol reference — §17]:

1. **supportedAppProtocolReq/Res** — agree protocol version (DIN 70121 vs ISO 15118-2 …).
2. **SessionSetupReq/Res** — session ID assigned.
3. **ServiceDiscovery / PaymentServiceSelection / PaymentDetails / Authorization** — services & authentication (certificates if Plug & Charge).
4. **ChargeParameterDiscoveryReq/Res** — exchange of EV limits (max V, max I, energy request) and EVSE limits (min/max V, max I).
5. **CableCheckReq/Res** — EVSE performs and reports insulation check of the cable; repeated until EVSE_IsolationStatus valid.
6. **PreChargeReq/Res** — EVSE raises output to battery voltage; EV confirms ΔV acceptable.
7. **PowerDeliveryReq/Res (Start)** — contactor closure; energy flow begins.
8. **CurrentDemandReq/Res** — cyclic (~per 10 ms-class tick in many implementations; timing is version/vendor-dependent) loop: EV sends requested V/I; EVSE replies with actual V/I, EVSE limits, and status.
9. **PowerDeliveryReq (Stop)**, **WeldingDetection**, **SessionStop** — controlled ending.

Fig. 7 — EV ↔ Charger Communication Sequence (DC session, DIN 70121 / ISO 15118)

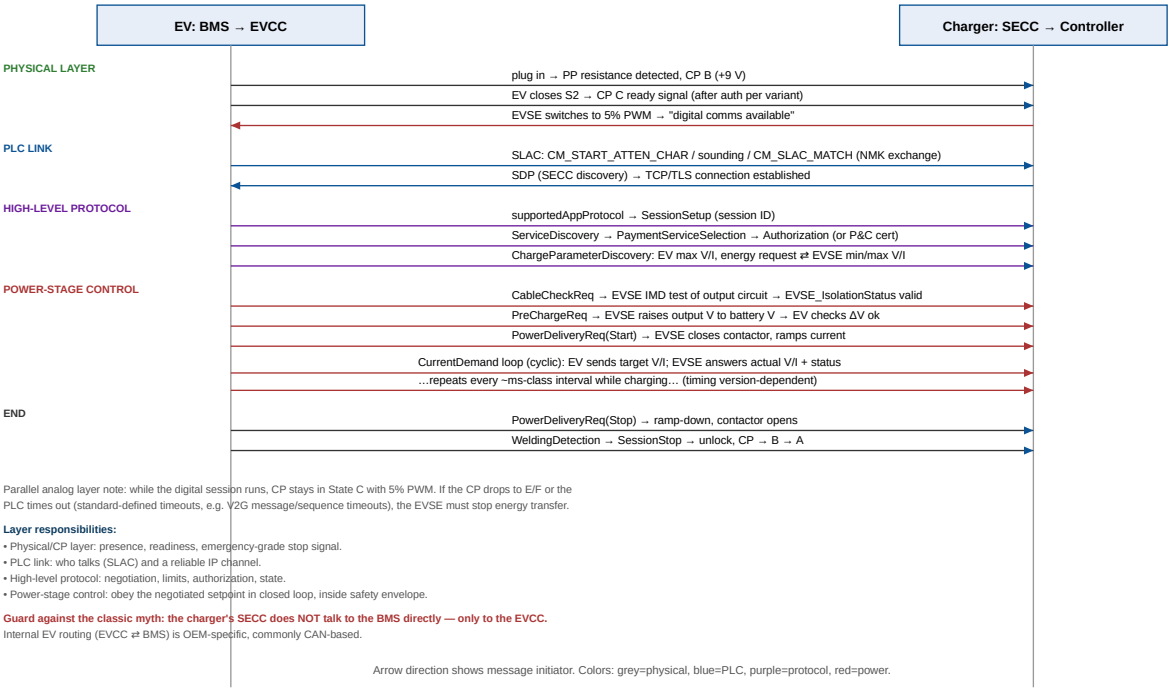


Figure 7 — Original communication sequence diagram. Message names per ISO 15118-2 reference [Typhoon HIL, §17]; SLAC primitives per ISO 15118-3.

5.6 Limits, status, timeouts, security

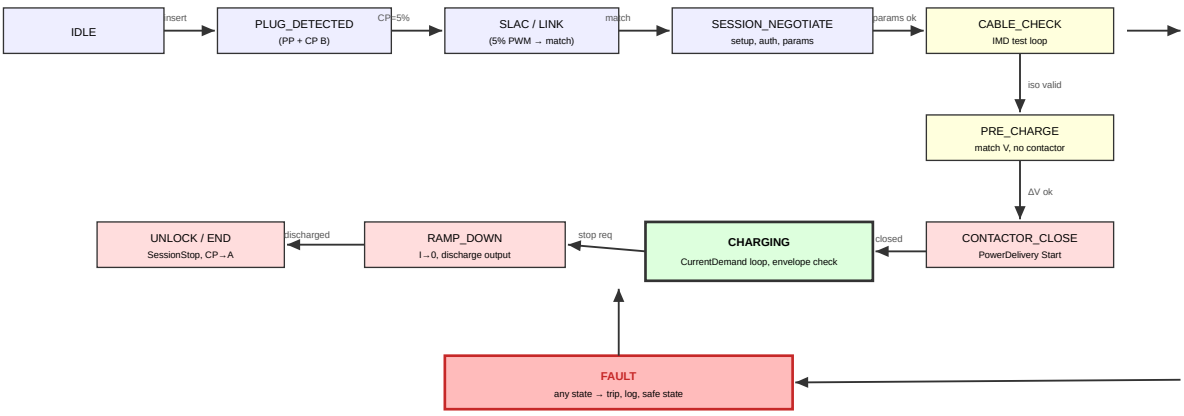
- **Charging limits:** EV sends EVMaximumVoltageLimit / EVMaximumCurrentLimit (and target values); EVSE replies with its own EVSEMaximum* — the operating point must satisfy both envelopes; both also carry temperature-based derating inputs.
- **Session start/stop:** started by Authorization + PowerDelivery(Start); stopped by EV request, EVSE decision, or timeout — always via PowerDelivery(Stop) + SessionStop.
- **Timeouts:** the standards define message-level and sequence-level timeouts (e.g. V2G_SECC_Sequence_Timeout, message timeouts per message family). Precise values are normative in DIN/ISO tables — treat exact numbers as "check the standard", not memorised trivia [Source: OSTI timeouts best-practices paper, §17].
- **Security:** DIN 70121: none built-in (external auth). ISO 15118-2: optional TLS + Plug & Charge certificate chain (contract cert ↔ OEM provisioning cert ↔ V2G root). ISO 15118-20: TLS mandated, bidirectional energy, refined P&C [Source: iTeh 15118-20 abstract; Porter 2025 paper, §17].

> **Architecture/version dependence.** Many vehicles on the road today still use DIN 70121; some newer ones use ISO 15118-2; ISO 15118-20 support is only beginning to appear. An SECC must support several stacks and pick one during supportedAppProtocol. Whether authentication happens on the charger screen, via app/RFID, or via P&C depends on protocol version and operator.

6. Charging Sequence: Step-by-Step

Stage	What happens	Who acts	Guard/verify
1. Cable connection	User plugs gun; PP resistance changes; inlet wakes	User + EV	PP plausibility
2. Connector & safety checks	Charger verifies earth, CP hardware path, lock actuation, cable temp	Charger	Ground OK, temp OK
3. CP state detection	CP B (+9 V) seen; later EV closes S2 → C	Both	Diode + level check
4–5. PLC link + SLAC matching	5% PWM; sounding/attenuation matching; NMK set; TCP(+TLS) up	EVCC/SECC	SLAC match to own gun only
6–8. Protocol & parameter negotiation	supportedAppProtocol → SessionSetup → auth → ChargeParameterDiscovery (limits both ways)	EVCC ⇄ SECC	Envelope intersection
9. Insulation & safety checks	IMD tests DC cable; EVSE_IsolationStatus valid	Charger IMD	Warning vs fault thresholds
10. Pre-charge / voltage matching	Output raised to battery V; EV accepts when ΔV within its limit	Charger + EV	No contactor closure on mismatch
11. Contactor operation	PowerDelivery(Start); contactor closes; current ramps	Charger	Aux-contact feedback; no weld
12–14. Controlled delivery, monitoring, dynamic adjustment	CurrentDemand loop: closed-loop V/I control; BMS may step current down (CCL) as SOC rises or temperature demands	Both	Continuous envelope supervision
15–16. Stop request, ramp-down, disconnection	PowerDelivery(Stop); ramp to 0; discharge output; contactor opens; WeldingDetection; unlock	Both	Output below safe threshold before unlock
17. Fault handling / termination	Any abnormality → immediate stop path; log, derate or trip per severity	Charger	Failsafe: stop energy on silence

Fig. 8 — Charger Session State Machine (educational)



Pseudocode (educational only — NOT production safety firmware):

Figure 8 — Original charging state-machine diagram.

```
state = IDLE
loop forever:
  switch state:
    IDLE:          if PP_present() and CP_read()==B: state = PLUG_DETECTED
    PLUG_DETECTED: enable_5pct_PWM(); start_SLAC()
```

```
                if SLAC_matched() and TCP_TLS_up(): state = SESSION
  SESSION:      run DIN/ISO messages; if params_exchanged(): state = CABLE_CHECK
  CABLE_CHECK:  IMD_test(); if isolation_valid(): state = PRECHARGE
  PRECHARGE:    set_output_V(battery_V)  // soft, via DC/DC, contactor still open
                if EV_confirms_deltaV_ok(): state = CLOSE
  CLOSE:        close_contactor(); aux_feedback_ok() or FAULT
                state = CHARGING
  CHARGING:     I_target = min(EV_request, charger_limit, thermal_limit)
                closed_loop regulate(V,I); monitor envelopes every loop
                if EV_stop or fault or timeout: state = RAMP_DOWN
  RAMP_DOWN:    ramp I→0; open_contactor(); discharge_output();
                welding_check(); unlock(); state = IDLE
  FAULT:        immediate: gates_off(), contactor_open(), log(), alert()
                state = IDLE   // after safe conditions met
watchdog_kick()
```

(ಕನ್ನಡ: ಸ್ಟೇಟ್ ಮಷೀನ್ ಎಂದರೆ — ಚಾರ್ಜರ್ ಒಂದೊಂದು ಹಂತದಲ್ಲಿ ಒಂದೇ ಒಂದು ಕೆಲಸ ಮಾಡುತ್ತದೆ: ಪ್ಲಗ್ ಪತ್ತೆ → ಸಂಭಾಷಣೆ → ಸುರಕ್ಷತಾ ಪರೀಕ್ಷೆಲನೆ → ವೋಲ್ಟೇಜ್ ಹೊಂದಾಣಿಕೆ → ಚಾರ್ಜಿಂಗ್ → ಸ್ಥಗಿತ. ಪ್ರತಿ ಹಂತದ ನಂತರ ಎಲ್ಲವೂ ಸರಿಯಾಗಿದೆಯೇ ಎಂದು ಪರೀಕ್ಷಿಲಿಸಿ ಮುಂದುವರುತ್ತದೆ; ಏನಾದರೂ ತಪ್ಪಿದರೆ FAULT ಸ್ಥಿತಿಗೆ ಹೋಗಿ ಸುರಕ್ಷಿತವಾಗಿ ನಿಲ್ಲುತ್ತದೆ.)

> Real certified chargers add redundant hardware paths (e.g. hardwired shutdown independent of the MCU), watchdog supervision, and interlocks. Treat the above as the mental model, never as implementable safety code.

7. Buck Converter: Full Technical Explanation

7.1 What and why

A **buck (step-down) converter** produces a lower average DC output voltage than its DC input, at higher current, using a switch, an inductor, a diode (or synchronous MOSFET) and an output capacitor. It is used wherever efficient voltage reduction is needed — far more efficient than a linear regulator because the pass element is either fully ON (low drop) or fully OFF, not dissipating the difference [Source: TI SLVA057 "Understanding Buck Power Stages" — §17].

Fig. 9 — Buck Converter: circuit and ON/OFF current paths

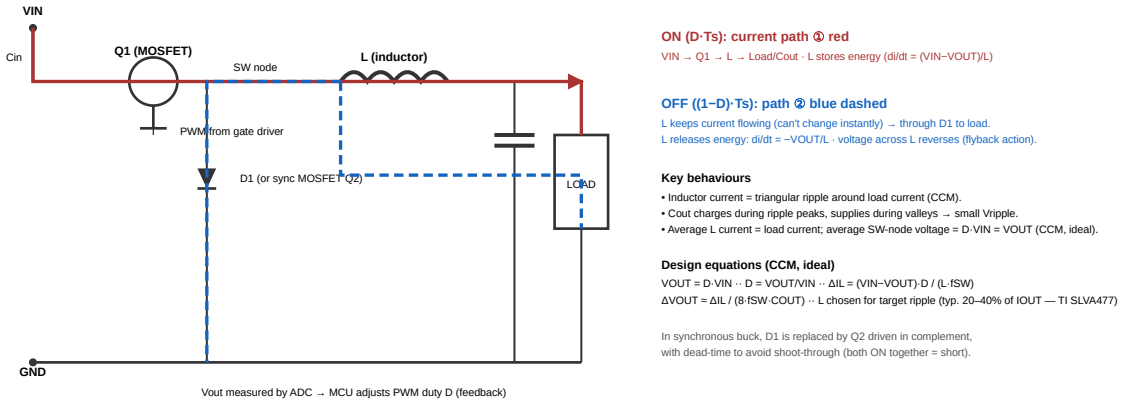


Figure 9 — Original buck converter circuit with ON (solid red) and OFF (dashed blue) current paths.

7.2 Switching cycle, PWM, and control

- **PWM duty cycle D:** fraction of the switching period T_s during which Q1 is ON. The MCU timer (ePWM) generates it; the gate driver amplifies it to drive Q1's gate.
- **ON phase:** input voltage appears across L minus output; inductor current ramps up at $(V_{IN}-V_{OUT})/L$; energy is stored in the magnetic field.
- **OFF phase:** the inductor *forces* current to continue (its voltage reverses), turning on D1; current ramps down at $-V_{OUT}/L$.
- **Volt-second balance** (average inductor voltage over a cycle = 0): $(V_{IN}-V_{OUT})\cdot D\cdot T_s = V_{OUT}\cdot(1-D)\cdot T_s$ →

$$V_{OUT} \approx D \cdot V_{IN} \quad \text{(ideal, continuous conduction mode)}$$

7.3 Worked example (stated assumptions)

Assumptions: ideal components (no $R_{DS(on)}$, no diode drop), CCM, f_{SW} = 100 kHz, V_{IN} = 24 V, V_{OUT} = 12 V, I_{OUT} = 5 A, target ripple 30% of I_{OUT}.

- $D = V_{OUT}/V_{IN} = 12/24 =$ **0.5**
- $\Delta I_L = (V_{IN}-V_{OUT})\cdot D/(L\cdot f_{SW})$ → choose L: with L = 33 µH, $\Delta I_L = 12 \times 0.5 / (33\ \mu \times 100\ k) \approx$ **1.82 A** (~36% of 5 A — matches the 20–40% design guideline [TI SLVA477, §17])
- Peak inductor current = 5 + 0.91 = **5.91 A** → sets saturation rating requirement
- Output ripple $\Delta V_{OUT} \approx \Delta I_L/(8\cdot f_{SW}\cdot C_{OUT})$: with C_{OUT} = 100 µF → ≈ **23 mV** (ideal estimate; ESR adds more)

7.4 Feedback control and the MCU's role

The controller samples V_{OUT} (and I_{OUT}) with the ADC, compares against the reference, and a compensator (typically PI/PID) adjusts duty D each cycle — a voltage-mode or current-mode loop. In embedded terms: **ADC trigger synchronised to PWM → control ISR → updated compare register**. Gate drivers provide the current amplification and galvanic isolation that an MCU output cannot. Losses to mention: conduction (I^2R), switching (V·I overlap), diode/conduction of freewheel, magnetic (core + copper), capacitor ESR. Thermal design and protection (OCP, OVP, thermal shutdown) follow [Source: TI SLVA057; Infineon gate-driver app notes — §17].

7.5 Relevance to EV chargers

Non-isolated high-power DC chargers (common in the 400 V class) use buck-derived output stages regulating battery current from the DC link; buck concepts also appear in auxiliary supplies, gate-driver bias rails, and the vehicle's own low-voltage DC/DC. In bidirectional chargers, the same power stage run in reverse performs boost operation — one reason topology choice matters (§9).

> *(ಕನ್ನಡ: ಬಕ್ ಕನ್ವರ್ಟರ್ ಎಂದರೆ— ಹೆಚ್ಚಿನ ವೋಲ್ಟೇಜನ್ನು ಕಡಿಮೆ ಮಾಡುವ ಸರ್ಕ್ಯೂಟ್. ಸ್ವಿಚ್ ಆನ್ ಮಾಡಿದಾಗ ಇಂಡಕ್ಟರ್ ಶಕ್ತಿ ಸಂಗ್ರಹಿಸುತ್ತದೆ; ಆಫ್ ಮಾಡಿದಾಗ ಅದು ಶಕ್ತಿಯನ್ನು ಲೋಡ್‌ಗೆ ಬಿಡುತ್ತದೆ. ಸ್ವಿಚ್ ಆನ್ ಇರುವ ಸಮಯದ ಪ್ರಮಾಣ (Duty Cycle D) ಬದಲಾಯಿಸಿ ಔಟ್‌ಪುಟ್ ವೋಲ್ಟೇಜನ್ನು ನಿಯಂತ್ರಿಸುತ್ತಾರೆ: Vout ≈ D × Vin.)*

8. Boost Converter: Full Technical Explanation

8.1 What and why

A **boost (step-up) converter** produces a higher average output voltage than its input, using the same component set in a different arrangement: L from input to switch node, switch to ground, diode to output [Source: TI SLVA061 "Understanding Boost Power Stages" — §17]. In EV chargers its most important role is the **PFC stage**: an active boost front end shapes grid current to be sinusoidal while raising rectified mains to a regulated DC-link voltage above peak line voltage.

Fig. 10 — Boost Converter: circuit and ON/OFF current paths

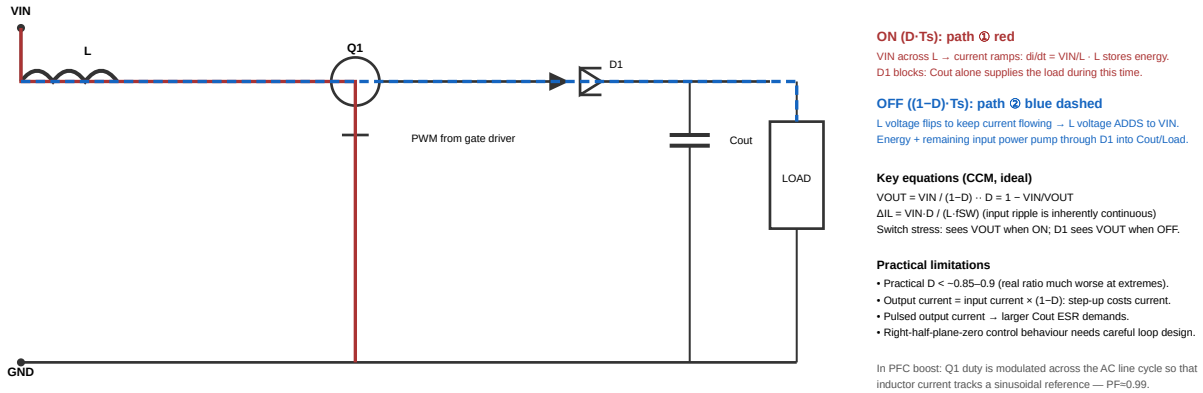


Figure 10 — Original boost converter circuit with ON (red) and OFF (dashed blue) current paths.

8.2 Operation, equation and worked example

ON: L charges from V_{IN} ($di/dt = V_{IN}/L$); D1 blocks because its anode is at ground. **OFF:** inductor voltage reverses to maintain current; the inductor voltage adds to V_{IN}, forward-biasing D1 and charging C_{OUT} to a voltage *above* the input. Volt-second balance: $V_{IN}\cdot D\cdot T_s = (V_{OUT}-V_{IN})\cdot(1-D)\cdot T_s \rightarrow$

$$V_{OUT} \approx V_{IN} / (1 - D) \quad \text{(ideal, CCM)}$$

Worked example. *Assumptions:* ideal, CCM, f_{SW} = 65 kHz, V_{IN} = 230 V DC (post-rectified single phase), target V_{OUT} = 400 V, P_{OUT} = 2 kW.

- $D = 1 - V_{IN}/V_{OUT} = 1 - 230/400 =$ **0.425**
- Input (inductor) current $\approx P/\eta/V_{IN}$: at $\eta = 0.97 \rightarrow \approx$ **8.96 A** average
- $\Delta I_L = V_{IN}\cdot D/(L\cdot f_{SW})$: with L = 500 µH $\rightarrow$ 230 × 0.425/(500 µ × 65 k) ≈ **3.0 A** peak-to-peak
- Peak switch current = $I_{L,avg} + \Delta I/2 \approx$ **10.5 A** $\rightarrow$ sets switch current rating and sense range

8.3 Feedback, sensing and protection

Closed-loop control senses V_{OUT} (voltage loop) and, in PFC operation, the input inductor current (current loop) — both into the MCU's ADC; the modulator updates duty. Protections: output overvoltage (loss-of-feedback danger — the load can see unregulated high voltage), switch overcurrent, and input undervoltage. Losses mirror the buck case plus higher diode stress; synchronous rectification and SiC devices reduce them.

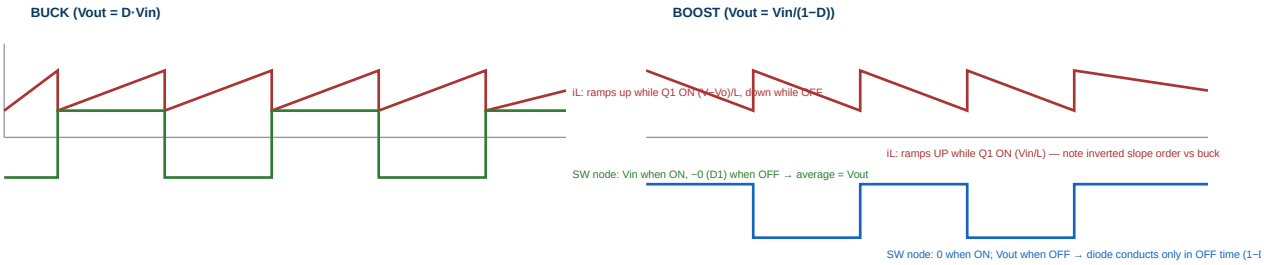
8.4 Relevance to EV chargers

- **Grid-side PFC boost rectifier** — the classic use: builds the ~800 V DC link from the AC mains while keeping power factor near unity.
- **Vehicle-side boost in V2G/bidirectional chargers** — energy can flow battery → DC link → grid, where the output stage boosts battery voltage up to the link.
- Classical (non-isolated) boost alone is unsuitable as the EV-facing stage in most DC chargers because isolation between grid and vehicle is mandated — isolation is provided by the DC/DC transformer stage, and the boost function appears inside or around it depending on topology.

> *(ಕನ್ನಡ: ಬೂಸ್ಟ್ ಕನ್ವರ್ಟರ್ ಎಂದರೆ — ಕಡಿಮೆ ವೋಲ್ಟೇಜನ್ನು ಹೆಚ್ಚಿಸುವ ಸರ್ಕ್ಯೂಟ್. ಸ್ವಿಚ್ ಆನ್ ಆದಾಗ ಇಂಡಕ್ಟರ್ ಕರೆಂಟ್ ಸಂಗ್ರಹಿಸುತ್ತದೆ; ಆಫ್ ಆದಾಗ ಇಂಡಕ್ಟರಿನ ವೋಲ್ಟೇಜು ಇನ್‌ಪುಟ್‌ಗೆ ಸೇರುತ್ತಾ ಹೆಚ್ಚಿನ ಔಟ್‌ಪುಟ್ ಕೊಡುತ್ತದೆ. Vout ≈ Vin / (1−D). ಚಾರ್ಜರ್‌ನ ಗ್ರಿಡ್-ಪಕ್ಕದ PFC ಹಂತದಲ್ಲಿ ಇದು ಬಹಳ ಮುಖ್ಯ.)*

9. Buck vs Boost vs Buck-Boost

Fig. 11 — Buck vs Boost: inductor current & switch-node waveforms (CCM, D = 0.5)



Reading the plots: the inductor ripple triangle is the same shape idea, but in the buck the ramp-up slope is (Vin−Vout)/L while in the boost the ramp-up slope is Vin/L during ON — the switch OFF-time conduction is what sets the conversion ratio.

Idealised CCM waveforms drawn for D=0.5, equal ripple slopes for clarity. Real designs are as per TI SLVA057/SLVA061.

Figure 11 — Original buck vs boost waveform comparison.

Aspect	Buck	Boost	Buck-Boost (inverting / 4-switch / SEPIC family)
Voltage behaviour	Steps down: $V_{OUT} < V_{IN}$	Steps up: $V_{OUT} > V_{IN}$	Any ratio, above or below; polarity-inverting classic or 4-switch non-inverting
Duty relation (CCM ideal)	$V_{OUT} = D \cdot V_{IN}$	$V_{OUT} = V_{IN}/(1-D)$	$\lvert V_{OUT}\rvert = D \cdot V_{IN}/(1-D)$ (classic inverting)
Circuit components	Series switch, L to output, freewheel diode/sync FET, C_{OUT}	L from input, shunt switch, series diode/sync FET, C_{OUT}	Two switches + L (4-switch) or coupled inductors (SEIC/Ćuk variants)
Input current	Pulsated (needs C_{IN})	Continuous (good EMI on input)	Depends on topology
Efficiency considerations	High; simple; good at moderate ratios	High but switch/diode see full V_{OUT}; harder at high ratios; RHP-zero loop	More switches → more conduction loss; 4-switch excellent for $V_{IN} \approx V_{OUT}$ ranges
Typical uses	Regulated lower rail from a bus; non-isolated charger output stages; aux supplies	PFC front end; DC-link generation; V2G reverse path	Battery-to-bus interfaces where battery voltage straddles the bus; USB-PD; LED drivers
Limitations	Cannot step up; isolation absent	Cannot step down; unregulated failure mode is dangerous (see §12)	Complexity/control effort; rarely used alone at high power

What real DC fast chargers actually use (architecture-dependent). A production DC fast charger is *not* "a buck or a boost". It is typically: an **active PFC front end** (boost-type, often in 3-phase bridgeless/Vienna or two-level AFE form) creating a DC link, followed by an **isolated DC/DC stage** — LLC, CLLC, DAB, or phase-shifted full-bridge — frequently **modular and interleaved/parallel** for power scaling and redundancy. Some designs are non-isolated buck-derived on the output (where applicable/allowed), and bidirectional (V2G) units use bidirectional capable topologies (DAB/CLLC). Topology choice depends on: voltage range (400 V vs 800 V packs), mandatory grid-to-vehicle isolation, power level, efficiency targets, billing metering point, safety/isolation strategy and product architecture. Presenting "the charger is a buck converter" in the interview would be a red flag — say "the charger *contains* converter stages whose principles are buck/boost physics applied in more complex topologies."

10. Control System and Embedded Firmware

10.1 Architecture and peripherals

Fig. 12 — MCU / ADC / PWM / Sensor feedback-control diagram

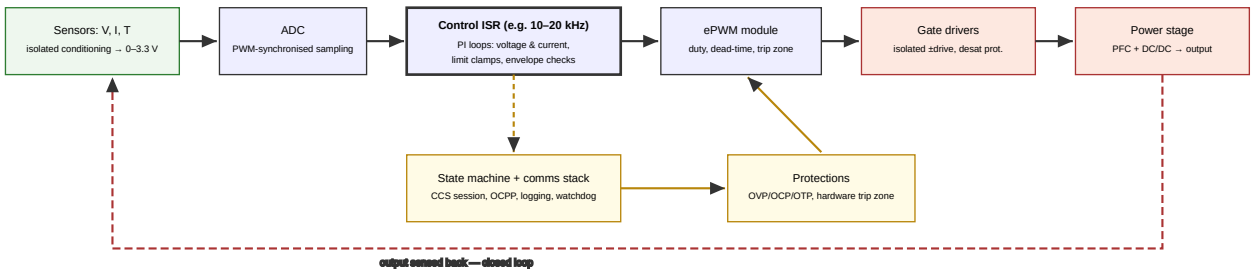


Figure 12 — Original feedback-control diagram: sensors → ADC → control ISR → PWM → gate drivers → power stage → back to sensors.

- **MCU architecture:** industrial MCUs/DSPs (e.g. C2000-class) with fast ADCs (12–16-bit), high-resolution ePWM with hardware dead-time and *trip zones* (hardware fault shutdown independent of software), and communication peripherals (CAN, UART/SPI/I²C, Ethernet with industrial stacks).
- **ADC sampling:** synchronised to PWM (sample at counter zero/period to average ripple or sample mid-ramp); multiple channels per cycle: output V, output I, DC-link V, temperatures, aux rails.
- **Feedback loops:** inner current loop (fast, kHz-class update), outer voltage/power loop, plus slower thermal derating loop. Anti-windup clamps; setpoint = min(EV request, charger capability, thermal limit, envelope).
- **State machine:** as in §6; implemented as an explicit enum-driven dispatcher — never a tangle of flags.
- **Communication stack:** SECC protocol runs on the main controller or a companion SoC; internal links on CAN (module controllers, sensors), RS-485/Modbus (meter), Ethernet (SECC, back-end).
- **Fault monitoring:** plausibility checks (twin sensors, physical consistency I·V≈P), timeout supervision on every link, hardware trip for OCP, watchdog with safe-state output.
- **RTOS vs bare-metal:** bare-metal super-loop + high-priority ISR is common for single-function controllers; RTOS (FreeRTOS, SafeRTOS, QNX-class) when session management, HMI, cloud connectivity and control must be cleanly partitioned; safety-certified parts of the design often isolate the critical loop in hardware/ISR regardless.
- **Event logging & debugging:** black-box flash logs (state transitions, faults, min/max, last N messages), JTAG/SWO trace in development, CAN/Ethernet diagnostics in field.
- **Calibration & testing:** ADC gain/offset calibration against reference instruments, dead-time verification, fault-injection (sensor disconnect, CP short, contactor weld simulation), EMC and environmental testing per IEC 61851-23 product standards.

> *(ಕನ್ನಡ: ಎಂಬೆಡೆಡ್ ಕಂಟ್ರೋಲರ್ ಪ್ರತಿ ಸೆಕೆಂಡಿಗೂ ಸಾವಿರಾರು ಬಾರಿ ಸೆನ್ಸರ್ ಮೌಲ್ಯಗಳನ್ನು ಓದಿ, ಕಾರು ಕೇಳಿದ ಕರೆಂಟ್‌ನೊಂದಿಗೆ ಹೋಲಿಸಿ, PWM ಅನ್ನು ಸ್ವಲ್ಪ ಬದಲಾಯಿಸುತ್ತದೆ— ಇದು ನಿರಂತರ ಸುತ್ತುವ "ಫೀಡ್‌ಬ್ಯಾಕ್ ಲೂಪ್". ಯಾವುದೇ ಅಸಂಗತತೆ ಕಂಡರೆ ತಕ್ಷಣ ಸ್ವಿಚ್ ಆಫ್ ಮಾಡುವ ಹಾರ್ಡ್‌ವೇರ್ ರಕ್ಷಣೆಯೂ ಇದೆ.)*

11. MATLAB/Simulink and Testing

11.1 Modelling the converters

Using Simscape Electrical, a buck converter is built from: a DC voltage source, MOSFET (or ideal switch), diode, series inductor, output capacitor, load, a *Controlled PWM Voltage* / *Half-Bridge Driver* pair for gate signals, and a PI controller comparing sensed output voltage to a reference. MathWorks' own worked example converts 30 V to a regulated 15 V and is explicitly intended for sizing L and C and designing the feedback controller; modelling MOSFETs (not ideal switches) captures on-resistance and switching timing, while an ideal-switch variant ("Buck Converter with Thermal Dynamics") trades fidelity for speed [Source: MathWorks Simscape Electrical Buck Converter example — §17].

Boost modelling recipe: same block set, rearranged per Fig. 10 — L between source and switch node, MOSFET to ground, diode to output; add the PI loop on V_{OUT}; for a PFC-flavoured exercise, replace the DC source with a rectified sinusoid and modulate duty across the cycle so inductor current tracks a sine reference.

11.2 Simulating feedback and studying behaviour

- **Voltage/current feedback:** discretise the controller at a realistic sample rate (e.g. 10–20 kHz), inject a load step, and watch the PI recover; log duty as well as V/I.
- **Duty sweeps:** run parameter sweeps of D and compare steady-state V_{OUT} with $D \cdot V_{IN}$ / $V_{IN}/(1-D)$ to validate the equations.
- **Ripple studies:** sweep L and C_{OUT}; measure ΔI_L and ΔV_{OUT} against the formulas from §7–§8.
- **Efficiency:** enable device loss models ($R_{DS(on)}$, diode drops, magnetics resistance); plot efficiency vs load.
- **State-transition testing:** model the session state machine as a Simulink Stateflow chart; drive it with scripted events (SLAC fail, CP drop, EV stop) and verify transitions with test assessments — this is exactly how protocol/state behaviour gets regression-tested.
- **Interpretation guide:** inductor-current triangle → check CCM/DCM boundary; output ripple → check ESR vs capacitive components; step-response overshoot → loop compensation quality; duty saturation → insufficient headroom (e.g. boost $D \to 1$).

11.3 HIL at a conceptual level

Hardware-in-the-loop replaces the physical power stage with a real-time simulator (e.g. Typhoon HIL-class, or Speedgoat with Simulink Real-Time): your real controller board drives PWM into the simulator, which emulates the converter and even a virtual EV/SECC/EVCC stack, returning sensed signals to your board. This lets teams test protection logic, fault reactions and protocol timeouts safely — before any HV hardware exists. The Typhoon HIL documentation explicitly supports EVCC/SECC ISO 15118-2 testing with Green PHY conversion modules [Source: Typhoon HIL docs, §17].

11.4 Beginner project plan (simulation-first)

Day	Task	Deliverable / proof
1	Build the MathWorks-style buck (24→12 V, 100 kHz). Validate $D \cdot V_{IN}$, ΔI_L, ripple vs formulas. Add PI loop; test a 50% load step.	Plots: V_{OUT} transient, iL ripple, duty trace; a table comparing measured vs calculated values
2	Repeat for boost (24→48 V). Try D = 0.9 and observe the ratio reality-check. Add current limiting and an OVP trip logic in the controller subsystem.	Efficiency vs load plot; trip-logic Stateflow chart with test results
3	Model the §6 charging state machine in Stateflow; simulate plug-in → pre-charge → charge → stop; inject CP loss and verify safe stop.	State transition log + short demo video/screenshot; summary of what you'd test on HIL next

> *(ಕನ್ನಡ: ಸಿಮ್ಯುಲಿಂಕ್‌ನಲ್ಲಿ ಮೊದಲು ಸರ್ಕ್ಯೂಟ್ ಅನ್ನು ಕಾಗದದ ಮೇಲೆ ಕಟ್ಟಿದ ಹಾಗೆ ಮಾಡಿ ಪ್ರಯೋಗ ಮಾಡಬಹುದು — ವೆಚ್ಚವಿಲ್ಲ, ಅಪಾಯವಿಲ್ಲ. ನಂತರ ಅದೇ ಲಾಜಿಕ್ ಅನ್ನು ನಿಜವಾದ ಬೋರ್ಡಿನಲ್ಲಿ HIL ಮೂಲಕ ಪರೀಕ್ಷಿಸುತ್ತಾರೆ.)*

12. Faults, Protection and Troubleshooting

Fig. 13 — Charger fault-detection flowchart (run each supervision cycle)

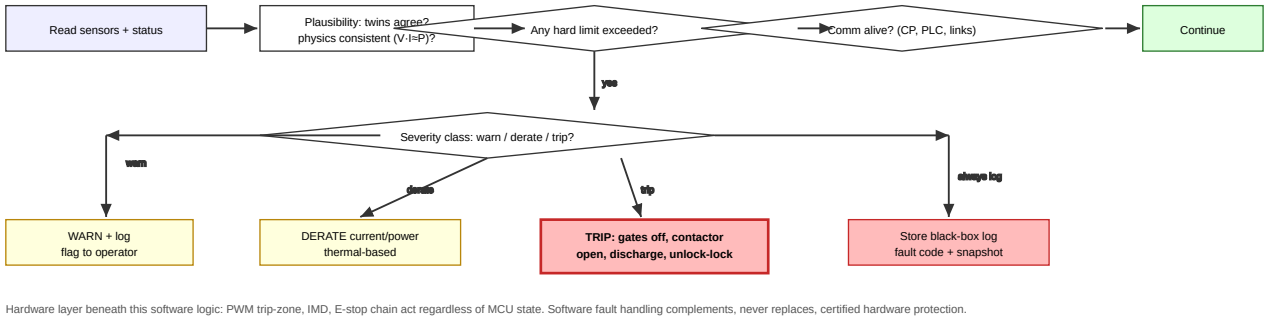


Figure 13 — Original fault-detection flowchart.

Fault	Symptom	Likely causes	Monitoring method	Typical safe response
Overvoltage	V above limit / EV stops	Loop instability, feedback loss, load dump	Redundant V sensing, HW comparator	Immediate gate-off, OVP latch, log
Overcurrent	I above EV request/limit	Setpoint error, sensor drift, short	Hall + shunt cross-check; HW OCP	Current limit then trip; hardware trip-zone
Overtemperature	Derate alarms; eventual stop	Cooling failure, fouled heatsink, loose lug	Multi-point NTC/PT100, plausibility vs twins	Progressive derating → stop; never bypass
Insulation fault	CableCheck fails; session refused	Damaged cable, moisture, EV-side isolation	IMD before + during session (IEC 61851-23 practice)	Refuse/stop energy transfer; only retry after diagnosis
Contactor failure	Commanded but no closure (or stuck closed)	Welded contacts, coil/driver fault	Aux contact feedback vs command	Block start; for weld — inhibit, require service
Pre-charge timeout	ΔV not reaching threshold in time	Excess EV-side capacitance, resistor open, leakage	Monitor V-match vs timer	Abort before contactor close; log energy/attempt
Communication timeout	Silence on PLC/CP	Modem fault, cable EMI, EV crash	Standards-defined sequence/message timers	Controlled stop within timeout; CP failsafe state
CP/PLC failure	No state transitions / no SLAC match	Dirty pins, wiring, diode monitor fault	CP level monitor; SLAC counters	Stay de-energised; never "force" charge
Sensor failure	Implausible/stuck values	Wiring, drift, reference loss	Twin sensors + physics checks	Fail-safe shutdown (missing data ≠ safe)
DC-link abnormality	Bus over/under-voltage, ripple rise	Cap aging, PFC fault, balance resistors	Bus V + ripple monitoring	Shutdown; capacitor condition inspection
Cooling failure	Flow alarm, rapid temp rise	Pump/fan failure, leak, clog	Flow switch, pump tach, ΔT	Derate/stop per thermal trajectory
Emergency stop	E-stop pressed / interlock open	Operator action, door switch	Hardwired chain into trip inputs	Immediate de-energise; requires manual reset

Diagnostic guidance vs certified safety design. The table describes how engineers *diagnose*; actual required response thresholds, timing and redundancy for a shipped product come from IEC 61851-23/-24 product certification, the EVSE safety concept and local electrical codes. Diagnostic insight never licenses bypassing a protection in the field.

13. Visuals and Images — Index of Original Diagrams

All figures in this report are **original drawings created for this report** (no copyrighted figures reproduced). Technical accuracy was prioritised; each carries a caption and is referenced from its section.

#	Diagram	Section	What it shows
Fig. 1	Overall DC fast charger architecture	§2	Power path, control path, communication path with all blocks
Fig. 2	Charger-side hardware block diagram	§3	One concrete board/cabinet-level arrangement + signal types
Fig. 3	EV-side hardware block diagram	§4	EVCC, BMS, pack, contactors, OBC context
Fig. 4	CCS2 connector pin/function illustration	§5.1	DC±, CP, PP, PE, AC pins and their roles
Fig. 5	CP/PWM concept diagram	§5.2	Pilot circuit concept + PWM levels + states
Fig. 7	EV–charger communication sequence diagram	§5.5	Layered sequence: physical → PLC → protocol → power
Fig. 8	Charging state-machine diagram	§6	IDLE→…→CHARGING→END with FAULT path
Fig. 9	Buck converter circuit + ON/OFF paths	§7	Schematic with coloured current paths and equations
Fig. 10	Boost converter circuit + ON/OFF paths	§8	Schematic with coloured current paths and equations
Fig. 11	Buck vs boost waveforms	§9	iL and switch-node waveforms compared
Fig. 12	MCU/ADC/PWM feedback-control diagram	§10	Closed loop through MCU peripherals and drivers
Fig. 13	Charger fault-detection flowchart	§12	Cycle of sense → classify → act
Fig. 6	PLC/SLAC/protocol-stack diagram	§5.3	Covered as the layered stack table in §5.3–5.4 (merged into the sequence figure's layer legend)

Note on external imagery: no external/stock images were embedded, so no licensing risk attaches to this document. If you later add photos of connectors or hardware, use licensed sources and credit them.

14. Interview Preparation — 32 Questions with Answers

Beginner (B), Intermediate (I), Scenario (S). Answers are compressed model answers — expand with your own words and the report's details.

B1. How does an EV and a DC charger communicate?

Two channels: (1) the analog **Control Pilot** — PWM states A–F per IEC 61851-1, where 5% duty signals "digital communication"; (2) a **digital PLC link over the CP line** using HomePlug Green PHY, established via SLAC, carrying DIN 70121 or ISO 15118-2/-20 messages between the EV's EVCC and the charger's SECC. The BMS's limits reach the charger via the EVCC, not directly.

B2. What is the charger controller's role?

It is the safety and regulation brain: runs the session state machine, converts the EV's negotiated V/I request into PWM setpoints for the power stage, supervises sensors and protections (IMD, temps, contactors), handles user interface/authorisation and back-end (OCPP) reporting, and guarantees safe shutdown on any fault or communication loss.

B3. What hardware exists on the charger side?

Answer with the three chains (see §3 table): power chain (breaker/SPD → EMI filter → PFC rectifier → DC-link caps → isolated DC/DC modules → contactors/pre-charge → CCS2 connector); control chain (MCU, gate drivers, V/I/T sensors, IMD, aux PSU); communication/user chain (CP/PP circuitry, PLC modem + SECC, meter, HMI/RFID, OCPP connectivity, cooling, E-stop).

B4. What is the difference between CP and PP?

CP is the bidirectional pilot: analog PWM states (presence/readiness/current availability/digital signalling). **PP** is proximity: detects that the plug is physically inserted and codes the cable's current-carrying capacity via a resistor — it doesn't "talk", it indicates.

B5. What is PLC in CCS2 charging?

Power Line Communication: data modulated onto the CP wire in the 2–30 MHz band per HomePlug Green PHY, so no extra pins are needed for digital messaging. It coexists with the PWM on the same conductor through coupling networks.

B6. What is SLAC?

Signal Level Attenuation Characterisation (ISO 15118-3): the EV sends sounding messages; the charger measures attenuation on its own CP and matches the vehicle whose attenuation profile belongs to *its* plug — establishing a unique secured PLC network (NMK key) between that one EV and that one gun.

B7. What are ISO 15118 and DIN 70121?

DIN SPEC 70121: the first deployed DC high-level protocol (no TLS, no Plug & Charge). ISO 15118-2: international successor adding Plug & Charge and optional TLS. ISO 15118-20: generation two — mandated TLS, bidirectional V2G energy transfer, richer services. An SECC typically supports all three and negotiates during supportedAppProtocol.

B8. What happens during charging negotiation?

SessionSetup → service/payment selection & authorisation → ChargeParameterDiscovery: EV presents max/target V, max I, energy need; EVSE presents its min/max V and max I; the operating point is the intersection. Then CableCheck (insulation) and PreCharge (voltage matching) before PowerDelivery(Start).

B9. What is a buck converter?

A step-down switching converter: a series switch chops the input; the inductor stores energy while the switch is ON and releases it through the freewheel diode while OFF; the output LC filters it. In CCM, $V_{OUT} = D \cdot V_{IN}$. Efficient because the pass device is never in its linear region.

B10. What is a boost converter?

A step-up switching converter: the shunt switch charges the inductor from the input while ON; when OFF, the inductor's reversed voltage adds to the input and pumps energy through a series diode into the output capacitor. CCM: $V_{OUT} = V_{IN}/(1-D)$. Found in charger PFC front ends.

B11. How does PWM control output voltage?

The load sees the time-average of the switched waveform. Changing duty cycle D changes that average (buck: linearly with D; boost: with 1/(1−D)). In closed loop, an MCU adjusts D from sensed V/I every control cycle.

B12. What is a gate driver?

A power amplifier/isolator between the MCU's 3.3 V PWM and the power switch's gate: provides the peak gate current for fast switching, galvanic isolation for high-side/floating switches, level shifting, and usually protection inputs (desaturation/OCP, Miller clamp, soft-off).

B13. Why are voltage and current sensors needed?

Closed-loop regulation (output must track the EV's request), envelope enforcement (never exceed EV/EVSE limits), billing (energy), plausibility (two sensors must agree), and protection (OVP/OCP hardware trips need fast, independent measurement).

B14. What is a DC link?

The intermediate DC bus between the grid-side rectifier and the DC/DC output stage, buffered by capacitors. It decouples the two conversion stages, lets each be optimised, and its voltage is a key health/status signal (also the dangerous stored-energy element that must be bled after shutdown).

B15. Why are contactors and pre-charge circuits used?

Contactors give a visible, physical isolation of the live output. Pre-charge limits the inrush current that would otherwise occur when connecting a "cold" charger output (or EV inverter capacitors) to a source — without it, contacts weld and components stress. Voltage matching (charger output ≈ battery) must be verified before the contactor closes.

B16. What is insulation monitoring?

Because a DC charger's output is isolated from earth, an insulation fault could be silent until someone touches the vehicle. An IMD continuously measures resistance between DC+ / DC− and PE, before and during the session, per IEC 61851-23 practice, refusing or stopping transfer below thresholds.

B17. What happens if communication is lost?

Each side runs standard-defined sequence/message timeouts; on expiry the EVSE de-energises in a controlled way (ramp-down, contactor open, discharge) and the CP fails to a no-power state. "Loss of communication = loss of energy" is the design axiom.

B18. What is a charger state machine?

An explicit set of session states (IDLE, PLUG_DETECTED, LINK, NEGOTIATE, CABLE_CHECK, PRECHARGE, CHARGING, RAMP_DOWN, FAULT) with guarded transitions — guaranteeing that every safety precondition is verified before energy flows, and that every exit path is safe.

B19. How can MATLAB/Simulink help?

Simscape models of the power stages validate equations, ripple and control loops without hardware; Stateflow models of the session state machine allow protocol/fault-path regression tests; and HIL rigs run your real controller against simulated power stages and virtual EV/SECC stacks — exactly what this company does (see §11).

B20. How would you debug a charging failure?

Structure: (1) reproduce and capture logs (charger event log, session IDs, fault codes); (2) identify the layer — physical/CP, PLC link, protocol, or power stage (the layered model of §5 makes this a decision tree); (3) measure — CP scope, SLAC counters, output V/I traces; (4) one-variable-at-a-time testing, including cross-checks with a known-good vehicle/charger; (5) never bypass protections to "make it work".

B21. How does embedded C/C++ fit into a charger controller?

C/C++ implements the control ISRs (PI loops, ADC-PWM synchronisation), the session state machine, protocol stack integration (often a certified SECC stack library), drivers (CAN, UART, SPI, Ethernet), logging/calibration frameworks, and safety monitors — with MISRA-style discipline, watchdog integration and unit tests for safety-relevant modules.

B22. EV-to-charger vs charger-to-backend communication?

EV ↔ charger (CCS): real-time, safety-critical, millisecond-class cyclic control, analog CP + PLC, DIN/ISO protocols. Charger ↔ backend (OCPP): non-real-time, operational — authorisation, billing, remote management, smart-charging profiles, over WebSocket/JSON (OCPP 1.6/2.0.1). They never mix: OCPP cannot talk to a car.

I23. Why does the CP switch to 5% duty cycle for DC charging?

Per IEC 61851-1's duty coding, 5% is the reserved code meaning "digital communication is required" — the EV then knows to start SLAC/ISO messaging rather than infer current from duty. (Flag: exact normative table is in the standard; this is the well-known summary.)

I24. Why is galvanic isolation needed, and where?

The grid-side and the vehicle-touchable output must be isolated for safety; this is achieved by the isolated DC/DC transformer stage. Isolation is also why output voltage can "float" relative to earth — and hence why IMD monitoring is mandatory in DC EVSE.

I25. Why are SiC MOSFETs attractive in chargers?

Higher blocking voltage, lower switching losses at high frequency → smaller magnetics, higher efficiency and power density — valuable in the PFC and DC/DC stages of high-power chargers. Trade-offs: gate-drive sensitivity (tighter dv/dt, gate-protection needs), cost, and EMI design care [Source: Infineon CoolSiC/EiceDRIVER material — §17].

I26. What does the CableCheck message actually do?

During CableCheckReq/Res, the EVSE runs its IMD test on the (contactor-closed) output cable circuit and reports EVSE_IsolationStatus; it may also be reporting its readiness to pre-charge. The EV waits for a valid status before requesting pre-charge — the digital confirmation of the analog-era safety check.

I27. What is the difference between CableCheck and PreCharge?

CableCheck proves the output circuit is safe (insulation). PreCharge raises the EVSE output to (approximately) the battery voltage with the contactor closed but current ~zero, so that when energy transfer begins, there is no voltage step that would cause inrush — the EV confirms the match.

I28. What is the CurrentDemand loop?

The cyclic core of a DC session: every interval, the EV sends its requested/target voltage and current (based on live BMS state); the EVSE replies with the actual delivered V/I and its own dynamic limits and status. This is how charging current tapers as SOC rises or temperatures change.

I29. What is WeldingDetection?

An ISO 15118/DIN message where, after PowerDelivery stop, the EV checks that the EVSE's output is de-energised — corroborating the mechanical/aux-contact weld checks — before unlocking and releasing the connector.

I30. What is OCPP and which version would you expect?

Open Charge Point Protocol (Open Charge Alliance): charger ↔ central-system messaging for authorisation, transactions, diagnostics, smart charging. 1.6 (SOAP/JSON-WS) is still widespread; 2.0.1 adds device management, improved transactions, security and ISO 15118 support; 2.1 extends to bidirectional/DER use [Source: Open Charge Alliance — §17].

S31. The charger detects the EV but charging never starts. Where do you look first?

Follow the layer ladder: CP states correct through C? If yes, did the 5% PWM appear? Then SLAC — did the match occur (modem logs)? Then protocol — which message failed (Authorization? ChargeParameterDiscovery? CableCheck isolation status?)? Each

layer's logs pinpoint the stage; most "no start" cases die at SLAC or authorization.

S32. Output voltage follows the request at low power but sags at high current. Reasoning?

Load-dependent sag suggests a real electrical cause: cabling/contactor drop (measure mV across joints), current-sensor gain error (calibration check vs reference meter), control-loop headroom (duty saturation? thermal derating active?), or DC-link collapse under load. Compare commanded vs measured vs independently metered values to separate sensing from power-path issues.

15. Practical Scenarios — Diagnostic Reasoning

Method for every scenario: (1) safety first — assume energised; (2) gather evidence (logs, LED/HMI codes, measurements); (3) locate the layer (physical/CP → PLC → protocol → power); (4) one change at a time; (5) document. "Documentation" includes the charger's commissioning file, protocol logs and the EV's known behaviour.

Scenario	Logical diagnostic sequence
1. Charger detects EV but charging does not start	Check CP state progression (B→C?) → verify 5% PWM present → SLAC match logs → protocol log: last message before stall (Authorization? params?) → IMD status → check user-side authorization (RFID/app). Classify: physical, link, protocol, or commercial.
2. PLC link cannot be established	Inspect CP coupling components and connector pins (oxidation) → check for interference sources and cable damage → try a second vehicle/gun to isolate EVCC vs SECC → verify modem firmware and attenuation margins; capture with a PLC sniffer if available.
3. Output voltage does not follow requested value	Compare EV request vs controller setpoint vs measured output at three points (controller, terminal, meter) → check feedback sensor calibration → check for duty saturation / thermal derate flags → inspect power path for voltage drops (contactor, lugs) under load.
4. Current readings are unstable	Determine: real ripple or measurement noise? Scope the sensor output vs a clamp meter → check sensor supply and shielding/grounding → verify ADC sampling sync with PWM → if real: control-loop oscillation (tuning), or EV-side oscillation (BMS request jitter) — logs will show request stability.
5. The converter overheats	Thermal imaging under controlled load → verify coolant flow/fan operation and filter condition → check I·V operating point vs design envelope (chronic overload?) → inspect thermal interface mounting → confirm derating logic engaged before limits (protection worked?).
6. The contactor does not close	Command present at driver? Coil voltage/current? Aux contact feedback? If commanded but silent: driver/coil/wiring; if closes but no feedback: aux contact or wiring; if mechanically jammed: replace (never force). Check interlocks upstream (E-stop chain) before concluding.
7. Pre-charge does not complete	Watch V-match trace: output rising at all? If flat: pre-charge resistor/relay path open. If rising but slow: excess EV-side capacitance or leakage (IMD reading?). If it reaches target but EV rejects: ΔV threshold or EV-side logic — use both sides' logs.
8. Charging stops unexpectedly	Pull black-box log: fault code? If comm timeout: check modem health/EMI. If thermal: verify sensor plausibility. If EV-initiated: BMS derate/stop — normal behaviour near SOC or temperature limits. Rule out site-level (breaker, grid dip) via upstream logs.
9. EV requests less current than expected	This is often *correct* behaviour: BMS tapers by SOC, temperature, or cell limits — compare request vs BMS-predicted curve. If anomalous: check battery temperature and SOC data in the CurrentDemand log; check for EV-side derating flags; do not "fix" by overriding the EV.
10. A sensor reports an implausible value	Verify with an independent instrument at the harness → check twin-sensor disagreement → inspect wiring/references → decide per safety concept: implausible = untrustworthy = fail-safe, not "average it out". Log, replace, and recalibrate; confirm plausibility logic catches it next time.

> *(ಕನ್ನಡ: ಯಾವುದೇ ಸಮಸ್ಯೆ ಬಂದಾಗ ಮೊದಲು ಲಾಗ್‌ಗಳನ್ನು ನೋಡಿ, ನಂತರ ಅಳತೆ ಮಾಡಿ — ಊಹೆ ಮಾಡಬಾರದು, ರಕ್ಷಣಾ ವ್ಯವಸ್ಥೆಗಳನ್ನು ಕಡೆಗಣಿಸಬಾರದು. ಒಂದೊಂದೇ ಬದಲಾವಣೆ ಮಾಡಿ ಪರಿಣಾಮ ಗಮನಿಸಬೇಕು.)*

16. Glossary and Learning Roadmap

16.1 Abbreviations

Abbrev	Expansion	Abbrev	Expansion
AC	Alternating current	IMD	Insulation Monitoring Device
ADC	Analog-to-Digital Converter	MCU	Microcontroller Unit
AFE	Active Front End	NMK	Network Membership Key (HPGP)
BMS	Battery Management System	OBC	On-Board Charger
CCS	Combined Charging System	OCPP	Open Charge Point Protocol
CCM/DCM	Continuous / Discontinuous Conduction Mode	OVP/OCP/OTP	Over-Voltage/Current/Temperature protection
CP / PP	Control Pilot / Proximity Pilot	P&C	Plug & Charge
DC	Direct current	PFC	Power Factor Correction
DAB	Dual Active Bridge (converter)	PLC	Power Line Communication
DCFC	DC Fast Charger	PWM	Pulse-Width Modulation
EMI/EMC	Electromagnetic Interference / Compatibility	RTOS	Real-Time Operating System
EVCC / SECC	EV / Supply-Equipment Communication Controller	SLAC	Signal Level Attenuation Characterisation
EVSE	Electric Vehicle Supply Equipment	SPD	Surge Protection Device
HIL	Hardware-in-the-Loop	SOH / SOC	State of Health / State of Charge
HPGP	HomePlug Green PHY	V2G	Vehicle-to-Grid
HV	High Voltage	VCU	Vehicle Control Unit

16.2 Three-day focused study plan (prioritised on the manager's topics)

Day	Focus	Activities	Self-test gate
1 System + protocols	Big picture + communication (the manager's topics 1–3)	Morning: read §1, §2, §3; redraw Fig. 1 from memory. Afternoon: §5 — trace Fig. 7 sequence while naming each message; read the CharIN/ISO summaries. Evening: write your own 60-second "how EV and charger talk" explanation aloud.	Can you explain CP states, 5% PWM, SLAC, EVCC/SECC, and the session message order without notes?
2 Power conversion	Buck/boost + hardware depth (topics 4–5)	Morning: §7–§9 with Fig. 9–11; redo both worked examples on paper. Afternoon: build the MathWorks buck model; add a load step. Evening: §12 fault table — memorise the 5 most-likely faults and responses.	Can you derive $V_{OUT}=D\cdot V_{IN}$ and $V_{IN}/(1-D)$ and explain ripple and duty physically?
3 Integration + practice	Firmware, testing, interview drills	Morning: §6, §10 — write the state machine pseudocode yourself; sketch Fig. 12. Afternoon: §14 — answer all 32 questions aloud, record and review; §15 scenarios once more. Evening: skim §11 HIL content and prepare 2 questions to ask the manager.	Full mock interview: 10 random questions from §14 answered in under 2 minutes each.

17. References

All URLs accessed and verified on **10 October 2026**. Standards documents (IEC/ISO/DIN) are copyrighted; this report cites freely accessible summaries and documentation instead of reproducing their text, and marks exact normative values as "consult the standard".

#	Source	Used for
1	eInfochips — "IEC 61851 Explained: EV Charging Standards & Safety Guide" — einfochips.com	IEC 61851 overview, CP/PP roles, duty-cycle concept (secondary)
2	Geocadra — "IEC 61851 Inspection Form — EV Charging Safety" — geocadra.com	CP states A–F with voltages (secondary summary)
3	HDT Electronic — "How does EVSE work? Control pilot and proximity contact signals" — hdt-electronic.com	CP function per IEC 61851-1 Annex A (secondary)
4	Pico Technology forum — "PLC — 5% duty cycle indicates digital communication" — picoauto.com	5% duty = digital signalling for off-board DC (secondary)
5	Typhoon HIL — "ISO 15118-2 Protocol" documentation — typhoon-hil.com	Full ISO 15118-2 DC message sequence; EVCC/SECC roles; TCP/TLS transport; HIL testing context
6	CharIN — "Guide ISO 15118 Service HPC1 V1.3" — charin.global	DIN 70121 vs ISO 15118-2 scope differences
7	ISO 15118-3 standard text (FHWA regulatory docket attachment) — downloads.regulations.gov	HomePlug Green PHY matching/SLAC sequence chart, attenuation concept
8	ISO 15118-2 standard text (FHWA regulatory docket attachment) — downloads.regulations.gov	Application-layer message set scope
9	iTeh — "EN ISO 15118-20:2022" abstract — standards.iteh.ai	ISO 15118-20 scope: TLS/TCP/IPv6, bidirectional V2G
10	OSTI/EV Watt — "Timeouts Best Practices" (2025) — osti.gov	Existence and treatment of ISO 15118-2 / DIN 70121 communication timeouts
11	ZeroPath — "Brief Summary: CVE-2025-12357 SLAC Protocol MITM" — zeropath.com	SLAC sounding/matching mechanics (security context)
12	Codico — "HomePlug Green PHY for Charging Stations" — codico.com	HPGP modules per ISO 15118 / DIN 70121 (component context)
13	Texas Instruments — SLVA057 "Understanding Buck Power Stages in Switchmode Power Supplies" — ti.com/lit/pdf/slva057	Buck operation, CCM equations, inductor ripple theory
14	Texas Instruments — SLVA477 "Basic Calculation of a Buck Converter's Power Stage" — ti.com/lit/pdf/slva477	20–40% ripple design guideline
15	Texas Instruments — SLVA061 "Understanding Boost Power Stages in Switchmode Power Supplies" — ti.com/lit/pdf/slva061	Boost operation, CCM equations, limitations
16	Infineon — Gate Driver ICs for SiC MOSFETs / EiceDRIVER — infineon.com	SiC gate-drive practice, isolation, protection features
17	MathWorks — "Buck Converter" Simscape Electrical example — mathworks.com	Simulink modelling recipe, controller design intent, ideal vs MOSFET switching models
18	MathWorks — "Buck Converter" block reference — mathworks.com	Controller-driven buck block semantics
19	Bender — "Electrical Safety for Electric Vehicle Charging" — benderinc.com	IMD requirement for DC EVSE per IEC 61851-23 practice
20	Gongyuan Global — "Insulation Monitoring Device for EV Charger (IEC 61851-23)" — global-gongyuan.com	IMD thresholds, pre-charge insulation check workflow
21	Open Charge Alliance — "OCPP 2.0.1" — openchargealliance.org	OCPP 2.0.1 scope: security, device management, smart charging
22	NREL — "The 2030 National Charging Network" — afdc.energy.gov	DCFC infrastructure context and scale
23	Texas Instruments — "AFE for Insulation Monitoring in High-Voltage EV Charging" (TIDUEZ8) — ti.com	IMD implementation reference design context

24	Wolfspeed — "What's Under the Hood: EV Chargers — A Tale of Standards" — wolfspeed.com	Connector/standards landscape, SiC relevance

Verified facts vs assumptions, in one place. Verified with citations above: CP states A–F; 5% duty = digital signalling; ISO 15118-2 message names and order; SLAC purpose; HPGP as PLC physical layer; TLS in 15118-2/mandatory in -20; DIN 70121 lacking P&C; OCPP 2.0.1 scope; IMD requirement practice for DC EVSE; buck/boost CCM equations and ripple guideline (TI); MathWorks buck modelling approach. **Illustrative/assumed values** (marked "ill." or "assumptions"): all example voltages, currents, frequencies and component values in worked examples; typical power-class figures; auxiliary rail voltages. **Normative tables** (duty-cycle mapping bands, timeout values, resistance codings) are summarised from secondary sources and flagged — always check IEC 61851-1/-23 and ISO 15118 parts for design values.

— End of report · Good luck with your interview! —